Post-Assessment

Question # 2 of 2

Limit points of set $S = \{(x, y) : -n < x, y < n, (x, y) \neq (0, 0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, –

Select the correct option

○	$\{(0,0)\} \cup \{(x,y) : x, y = \pm n, n \in \mathbb{N}\}$
◉	$\{(0,0)\} \cup \{(x,y) : x, y = \pm n, n \in \mathbb{N}\} \cup S$
○	$\{(x,y) : x, y = \pm n, n \in \mathbb{N}\} \cup S$
○	$\{(0,0)\} \cup S$

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ment Quiz

ost-Assessment

uestion # 2 of 2 **Time Left**

If H is an open covering of a compact subset S, then S can be covered by _____ many sets from H.

elect the correct option Reload Math Ed

- ○ infinitely
- ◉ finitely
- ○ non of these
- ○ both a and b

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Post-Assessment

Question # 2 of 2

Closure of the set $\{(x, y) : -n < x, y < n, (x, y) \neq (0, 0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is –

Select the correct option

○	$\{(x, y) : (x, y) \neq (0, 0)\}$
●	$\{(x, y) : -n \leqslant x, y \leqslant n, n \in \mathbb{N}\}$
○	$\{(x, y) : x, y \leqslant n, (x, y) \neq (0, 0), n \in \mathbb{N}\}$
○	$\{(x, y) : -n < x, y < n, n \in \mathbb{N}\}$

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Question # 1 of 2

A sequence of points $\{X_r\}$ in R^n converges to the limit $\bar{X}$ if _

Select the correct option

○	$\lim_{r\to\infty}\left\|X_r - \bar{X}\right\| = 1$
○	$\lim_{r\to 0}\left\|X_r - \bar{X}\right\| = \bar{X}$
●	$\lim_{r\to\infty}\left\|X_r - \bar{X}\right\| = 0$
○	$\lim_{r\to 0}\left\|X_r - \bar{X}\right\| = 0$

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Assessment Quiz

Post-Assessment

Question # 1 of 2

Exterior of the set $\{(x, y) : -n < x, y < n, (x, y) \neq (0, 0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is

Select the correct option

- ○ $\{(x, y) \in \mathbb{R}^2 | \, |x| > n, |y| < n, n \in \mathbb{N}\}$
- ◉ $\{(x, y) \in \mathbb{R}^2 | \, |x| > n, |y| > n, n \in \mathbb{N}\}$
- ○ $\{(x, y) \in \mathbb{R}^2 | \, |x| < n, |y| > n, n \in \mathbb{N}\}$
- ○ $\{(x, y) \in \mathbb{R}^2 | \, |x| < n, |y| < n, n \in \mathbb{N}\}$

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ost-Assessment

Question # 1 of 2

The $\lim_{X \to X_0} f(x) = -\infty$ if X_0 is a limit point of D_f and M there is a $\delta > 0$ such that _

Select the correct option

Reloc

- $f(x) \geq M \text{whenever} 0 < |X - X_0| \leq \delta$
- $f(x) < M \text{whenever} 0 < |X - X_0| < \delta$
- $f(x) \leq M \text{whenever} 0 < |X - X_0| \leq \delta$
- $f(x) > M \text{whenever} 0 < |X - X_0| < \delta$

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ment Quiz

ost-Assessment

uestion # 1 of 2

The function $f(x,y) = \frac{xy}{x^2+y^2}$ is defined everywhere in _____ except at _.

elect the correct option

○	none of these
○	$R^2,\ (1,-1)$
◉	$R^2,\ (0,0)$
○	$R^3,\ (1,-1,0)$

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Post-Assessment

Question # 1 of 2

The $\lim_{|X|\to\infty} f(x) = L$ If D_f is _____

Select the correct option

○	Finite
○	Zero
●	Unbounded
○	Bounded

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Post-Assessment

Question # 2 of 2 **Time Left: 75 s**

The limit of the function $f(x) = \frac{xy}{x^2+y^2}$ by letting (x, y)approach $(0, 0)$along the line $y = -x$ is _____.

Select the correct option

Reload Math Equati

- ○ $\frac{1}{2}$
- ○ Undefined
- ◉ $-\frac{1}{2}$
- ○ Finite

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Question # 1 of 2

If $f(x,y,z) = \cos(\frac{1}{x^2+2y^2+z^2})$, then __

Select the correct option

●	$\lim_{\lvert X\rvert\to\infty} f(X) = 1$
○	Not Exist
○	$\lim_{\lvert X\rvert\to\infty} f(X) = \infty$
○	$\lim_{\lvert X\rvert\to\infty} f(X) = 0$

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Question # 2 of 2

If $\lim_{x \to x_0} = f(x)$ exists, then it is _

Select the correct option

○	One
○	Finite
○	Zero
●	Unique

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Pre-Assessment

Question # 2 of 2

Exterior of the set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is – -

Select the correct option

○	$\{(x,y) \in \mathbb{R}^2 \mid \|x\| > n, \|y\| < n, n \in \mathbb{N}\}$
○	$\{(x,y) \in \mathbb{R}^2 \mid \|x\| < n, \|y\| > n, n \in \mathbb{N}\}$
●	$\{(x,y) \in \mathbb{R}^2 \mid \|x\| > n, \|y\| > n, n \in \mathbb{N}\}$
○	$\{(x,y) \in \mathbb{R}^2 \mid \|x\| < n, \|y\| < n, n \in \mathbb{N}\}$

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Boundary of the set $\{(x, y) : -n < x, y < n, (x, y) \neq (0, 0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is -

Select the correct option

○	$\{(x, y) \in \mathbb{R}^2 \mid x = \pm n, y = \pm n, n \in \mathbb{N}\} \cap \{(0, 0)\}$
○	$\{(0, 0)\}$
○	$\{(x, y) \in \mathbb{R}^2 \mid x = \pm n, y = \pm n, n \in \mathbb{N}\}$
◉	$\{(x, y) \in \mathbb{R}^2 \mid x = \pm n, y = \pm n, n \in \mathbb{N}\} \cup \{(0, 0)\}$

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Pre-Assessment

Question # 2 of 2 **Time Left: 73**

A sequence of points $\{X_r\}$ in R^n converges if and only if for each _____ there is an integer K such that ____ if $r, s \geqslant K$..

Select the correct option

Reload Math Equa

- (●) $\varepsilon > 0, \quad |X_r - X_s| < \varepsilon$
- ( ) $\varepsilon < 0, \quad |X_r - X_s| < \varepsilon$
- ( ) $\varepsilon \neq 0, \quad |X_r - X_s| > \varepsilon$
- ( ) $\varepsilon > 0, \quad |X_r - X_s| > \varepsilon$

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Question # 2 of 2 **Time Le**

If H is an open covering of a compact subset S, then S can be covered by _____ many sets from H.

Select the correct option

Reload Math E

- ○ both a and b
- ◉ finitely
- ○ non of these
- ○ infinitely

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ost-Assessment

Question # 1 of 2 **Tim**

In $\mathbb{R}^n$, an open n - ball of radius r about X_0 is $B_r(X_0) = \{X : |X - X_0| < r\}$, which contains - - -

Select the correct option

Reload M

- limit points of each of its points
- an ε - neighborhood of each of its points
- limit points of each of its boundary
- only its isolated point at center X_0

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Post-Assessment

Question # 2 of 2

The space R^n is _____.

Select the correct option

●	connected
○	both a and b
○	not connected
○	none of these

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Post-Assessment

Question # 1 of 2

Sequence in $\mathbb{R}^2$ defined as;

$$\{X_r\} = \left\{\left(\sum \frac{1}{r}, \sum \frac{1}{r^2}\right)\right\} - - - - -$$

Select the correct option

○	is oscillatory
○	converges to $(1, 0)$
○	converges to $(0, 0)$
◉	diverges

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ment Quiz

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Question # 1 of 2 Time

In $\mathbb{R}^n$, an open n - ball of radius r about X_0 is $B_r(X_0) = \{X : |X - X_0| < r\}$, then its closure is - -

Select the correct option

Reload Mo

- $\{X : |X - X_0| > r\}$
- $\{X : |X - X_0| \geqslant r\}$
- $\{X : |X - X_0| = r\}$
- $\{X : |X - X_0| \leqslant r\}$

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uestion # 2 of 2 **Time Left:**

In $\mathbb{R}^n$, if the line segment: $X = tX_2 + (1-t)X_1,\ 0 < t < 1$, joining X_1 and X_2 lies in $S_\varepsilon(X_0)$, *then* –

elect the correct option

Reload Math Ec

- $|X - X_0| < 1 - \varepsilon$
- (●) $|X - X_0| < \varepsilon$
- $|X - X_0| < 1 - t$
- $|X - X_0| < t$

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Post-Assessment

Question # 1 of 2

Which of the following is correct for vector value functions?

Select the correct option

- ○ $\lim_{u \to u_0} G_i(u) = L$
- ○ $\lim_{U \to U_0} g_i(u) = L_i$
- ◉ $\lim_{U \to U_0} G(U) = \overrightarrow{L}$
- ○ All of these

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Question # 1 of 2

The set for composite function is _____

Select the correct option

◉	$T = \{U : U \in D_G \text{and} \mathrm{G}(U) \in D_f\}$
○	$G = \{T : T \in D_U \text{and} \mathrm{G}(U) \in D_f\}$
○	$U = \{T : T \in D_U \text{and} \mathrm{G}(U) \in D_f\}$
○	$R = \{T : T \in D_U \text{and} \mathrm{G}(U) \in D_f\}$

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Post-Assessment

Question # 1 of 2 **Time Left: 80 s**

Suppose that $g_1, g_2, \ldots g_n$ are real valued functions defined on a subset T of $\mathbb{R}^m$, then the vector valued function G on T is defined by,

Select the correct option

Reload Math Equatio

- ○ $G(U) = (g_1(U), g_2(U), \ldots g_n(U)), U \in R^m$
- ○ $G(U) = (g_1(u), g_2(u), \ldots g_n(u)), U \in T$
- ● $G(U) = (g_1(U), g_2(U), \ldots g_n(U)), U \in T$
- ○ $G(U) = (g_1, g_2, \ldots g_n), U \in T$

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Question # 2 of 2 Time Left: 81 sec(s)

Suppose that U_0 is in T and is a limit point of T, G is continuous at U_0, and f is continuous at $X_0 = G(U_0)$, then $h = foG$ is continuous at _____.

Select the correct option

Reload Math Equations

- (●) U_0
- ( ) $G(U_0)$
- ( ) All of these
- ( ) T

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Post-Assessment

Question # 1 of 2

The function $f(x, y) = 1 - x^2 - 2y^2$ is continuous on ___

Select the correct option

○	$\mathbb{Z}$
○	$\mathbb{R}$
●	$\mathbb{R}^2$
○	$\mathbb{N}$

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Post-Assessment

Question # 2 of 2

Can we redefine the function $f(x, y) = \frac{xy}{x^2+y^2}$ to make it continuous at $(0, 0)$.

Select the correct option

- ○ Yes
- ◉ No

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Question # 1 of 2

If X_0 is in D_f and is a limit point of D_f, then we say that f is continuous at X_0 if ___

Select the correct option

Reloc

- (●) $\lim_{X \to X_0} f(X) = f(X_0)$
- ( ) $\lim_{X \to X_0} f(X) = 0$
- ( ) None of these
- ( ) $\lim_{X \to X_0} f(X) = 1$

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Question # 2 of 2 **Time Left: 7**

Consider the function $h(x,y) = \begin{cases} \sin\frac{\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}, x^2+2y^2<1 \\ 1, x^2+2y^2=1 \end{cases}$ then it is continuous on the _____

Select the correct option

Reload Math Equa

- ○ Parabola
- ◉ Ellipse
- ○ Hyperbola
- ○ Circle

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Post-Assessment

Question # 2 of 2

The $\lim_{X \to X_0} f(x) = \infty$ if X_0 is a limit point of D_f and M there is a $\delta > 0$ such that ___

Select the correct option

Reloc

- (●) $f(x) > M \text{whenever} 0 < |X - X_0| < \delta$
- ( ) None of these
- ( ) $f(x) < M \text{whenever} 0 < |X - X_0| < \delta$
- ( ) $f(x) \leq M \text{whenever} 0 < |X - X_0| \leq \delta$

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Post-Assessment

Question # 2 of 2

Let $f(x, y) = \frac{xy}{x^2+y^2}$ then the limit of f along the line $y = x$ as (x, y) approach $(0, 0)$ is __

Select the correct option

- ○ 0
- ◉ 1/2
- ○ undefined
- ○ 2

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ment Quiz

st-Assessment

uestion # 1 of 2

If X_0 is a point in R^n and r is a positive number, the open n-ball of radius r about X_0 is the set $B_r(X_0) = \{X| \ |X - X_0| _____\}$.

elect the correct option

- $\leq r$
- $> r$
- $< r$
- $\geq r$

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Assessment Quiz

Post-Assessment

Question # 1 of 2

If $f(x, y, z) = \cos\left(\frac{1}{x^2+2y^2+z^2}\right)$, then _____.

Select the correct option

- $\lim_{|X|\to\infty} f(X) = 1$
- $\lim_{|X|\to\infty} f(X) = \infty$
- $\lim_{|X|\to\infty} f(X) = 0$
- Not Exist

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re-Assessment

Question # 3 of 3

In $\mathbb{R}^2$, for the function $f(x, y) = x^2 + 2xy$, the $d_{(2,1)}f = -$

Select the correct option

○	$4dx + 6dy$
○	$2dx + 6dy$
●	$6dx + 4dy$
○	$6dx + 2dy$

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Pre-Assessment

Question # 1 of 3

For the function

$$f(x, y) = 3x^2y^3 + xy$$

in two variables,

$$f_{yx}(0, 0) = \ldots\ldots\ldots\ldots$$

Select the correct option

◉	1
○	-1
○	0
○	None of these

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Pre-Assessment

Question # 3 of 3 **Time Left: 81 sec(s)**

If $f_{x_i}(X)$ exists at every point of a set D, then it defines a function f_{x_i} on D. if this function has a partial derivative with respect to x_j on a subset of D, we denote the partial derivative by

Select the correct option

Reload Math Equations

○	$\frac{\partial^2 f}{\partial x_j x_i}$
●	All of these
○	$f_{x_i x_j}$
○	$\frac{\partial}{\partial x_j}(\frac{\partial f}{\partial x_i})$

Click to Save Answer & Move to Next Question

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Pre-Assessment

Question # 3 of 3 **Time Left: 77**

In $\mathbb{R}^n$, a function f is differentiable at X_0, $\Leftrightarrow \exists$ a linear function L in a way that $f(X) - f(X_0)$ can be approximated near X_0 by L satisfying;

Select the correct option

Reload Math Equat

- [ ] None of the above
- [ ] $L\left(\frac{X}{X_0}\right) = \frac{LX}{LX_0},\ |X_0| \neq 0, LX_0 \neq 0$
- [ ] $L(XX_0) = (LX)(LX_0)$
- [x] $L(X - X_0) = LX - LX_0$

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uestion # 2 of 3 **Time Left: 7**

If $z_0 = f(x_0, y_0)$ then $z(t) = f(x + \phi t, y + \phi t)$ represents a curve through (x_0, y_0, z_0) in the plane determined by the unit vectors $\hat{\Phi}$ and $\hat{k}$, then $\frac{\partial f(x_0, y_0)}{\partial \hat{\Phi}}$ is the - - - - - - - - (through)at (x_0, y_0, z_0).

elect the correct option

Reload Math Equ

- ○ Slope of surface z=f(x,y)
- ◉ Slope of straight line
- ○ Slope of the curve
- ○ Slope of the plane

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re-Assessment

Question # 3 of 3

If f is differentiable at X_0, then there - - - - - - - - - - linear function(s) L satisfying;

i) $f(X) - f(X_0) = L(X - X_0) + E(X)(|X - X_0|)$

ii) $\lim_{X \to X_0} E(X) = 0$

Select the correct option

- (●) exactly one
- ( ) are infinite multiple
- ( ) are finite multiple
- ( ) exists no

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Pre-Assessment

Question # 1 of 3 **Time Left: 81 sec(s)**

Suppose that f, f_x, f_y and f_{xy} exist on a neighborhood N of (x_0, y_0), and f_{xy} is continuous at(x_0, y_0). Then $f_{yx}(x_0, y_0)$_____.

Select the correct option

Reload Math Equations

- ○ Not Exists
- ◉ Exists

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Pre-Assessment

Question # 3 of 3 **Time Left: 8**

A function f is differentiable at $X_0 = (x_{10}, x_{20}, \ldots, x_{n0})$ if $X_0 \in D_f^0$ and there are constants $m_1, m_2, \ldots, m_n$ such that

$$\lim_{X \to X_0} \frac{f(X - X_0) - \sum_{i=1}^{n} m_i(x_i - x_{i0})}{X - X_0} = 1$$

-

Select the correct option

Reload Math Equ

- ○ True
- ◉ False

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Question # 1 of 3 **Time Left:**

In the partial derivative as a directional derivative the unit vector E_3 is defined by _____.

Select the correct option

Reload Math Eq

- ○ $(0, 0, -1, \ldots, 0)$
- ◉ $(0, 0, 1, 0, \ldots, 0)$
- ○ $(1, 0, 0, \ldots, 0)$
- ○ $(1, 0, 1, \ldots, 0)$

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st-Assessment

ıestion # 1 of 2 **Tir**

In $\mathbb{R}^3$, a function $f(x,y) = (xy)^3$ has the critical point at $(0,0)$, and all f_{x_i} exist $\forall\, 1 \leqslant i \leqslant n$ in the neighborhood of $(0,0)$, then f $---$at $(0,0)$.

lect the correct option Reload

- (●) neither maximum nor minimum
- ( ) minimum
- ( ) both maximum and minimum
- ( ) maximum

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Question # 2 of 2

In $\mathbb{R}^n$, a function f has the local extreme point in the neighborhood of $X = A$, and all f_{x_i} exist $\forall\ 1 \leqslant i \leqslant n$, then - - - - - - .

Select the correct option

○	$f_{x_i}(A) \neq 0\ \forall\ 1 \leqslant i \leqslant n$
○	$f(A) = 0$
●	$f_{x_i}(A) = 0\ \forall\ 1 \leqslant i \leqslant n$
○	$f(A) \neq 0$

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Post-Assessment

Question # 2 of 2

For a function of several variables $f(\mathbf{X})$, a point $\mathbf{X}_0$ is local extreme point if

Select the correct option

○	partial derivatives exists and equal to zero.
○	None of these
○	$f(\mathbf{X}) - f(\mathbf{X}_0)$ for all $\mathbf{X} \in D_f$.
◉	If $f(\mathbf{X}) - f(\mathbf{X}_0)$ does not change sign in $S_\delta(\mathbf{X}_0) \cap D_f$.

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ost-Assessment

Question # 2 of 2 **Time Lef**

In $\mathbb{R}^n$, a function f is said to have the local extreme point at $X = A$, if there exists a $\varepsilon > 0$, such that $f(X) - f(A)$ does not change the sign in - - - - - - - - .

Select the correct option

Reload Math E

- $D_f \cup S_\varepsilon(A)$
- the domian D_f of f
- ε - neighborhood $S_\varepsilon(A)$ of point of $X = A$
- $D_f \cap S_\varepsilon(A)$

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Identify the true statement(s)

elect the correct option

- (●) If $f_{x_i}(\mathbf{X}_0) = 0$ then f may or may not has a local extreme point at $\mathbf{X}_0$
- ( ) None of these
- ( ) the function $f(x, y) = x^3 + y^3$ has a local extreme point at origin.
- ( ) At every point where $f_{x_i}(\mathbf{X}_0) = 0$ function has local extreme point.

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Question # 2 of 2

In $\mathbb{R}^n$, a function f has the critical point at $X = A$, and all f_{x_i} exist $\forall\, 1 \leqslant i \leqslant n$ in the neighborhood of $X = A$, then f ------ has the local extreme at A.

Select the correct option

- ○ necessarily
- ◉ not necessarily

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st-Assessment

uestion # 1 of 2

A vector valued function $\vec{H} = (h_1, h_2, \ldots, h_n)$ is differentiable at $\vec{V_0} = (v_{1_0}, v_{2_0}, \ldots, v_{n_0})$, if each of the component $- - - - -$ is differentiable at V_0.

elect the correct option

Relo

- (●) function $h_i, 1 \leqslant i \leqslant n$
- ( ) v_{i_0}

Click to Save Answer & Move to

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uestion # 2 of 2

A vector valued function $\vec{H} = (h_1, h_2, \ldots, h_n)$ is differentiable at $\vec{V_0} = (v_{1_0}, v_{2_0}, \ldots, v_{n_0})$, if - - - - - - - - - of the component function $h_i, 1 \leqslant i \leqslant n$ is differentiable at V_0.

elect the correct option

Re

- ○ least one
- ◉ each

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Question # 2 of 2

For a function of several variables
$f(\mathbf{X})$

, a point
$\mathbf{X}_0$

is local extreme point if

Select the correct option

- (●) If $f(\mathbf{X}) - f(\mathbf{X}_0)$ does not change sign in $S_\delta(\mathbf{X}_0) \cap D_f$.
- ( ) $f(\mathbf{X}) - f(\mathbf{X}_0)$ for all $\mathbf{X} \in D_f$.
- ( ) None of these
- ( ) partial derivatives exists and equal to zero.

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nent Quiz

st-Assessment

uestion # 1 of 2

In $\mathbb{R}^n$, a function f is said to have the local minimum point at $X = A$, if the following inequlity holds $\forall\, X \in\ D_f \cap S_\varepsilon\,(A)$.

lect the correct option

- $f(X) \geqslant f(A)$
- $f(A) > f(X)$
- $f(X) > f(A)$
- $f(A) \geqslant f(X)$

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Question # 1 of 2

In $\mathbb{R}^3$, a function $f(x, y)$ is differentiable at $A = (\alpha, \beta)$, if $f(X) = f(A) + f_x(A)(x - \alpha) + f_y(A)(x - \alpha) + E(X)(|X - A|)$ such that -

Select the correct option

- ○ $\lim_{X \to A} E(X) = 1$
- ○ $\lim_{X \to A} E(X) = \infty$
- ◉ $\lim_{X \to A} E(X) = 0$
- ○ $\lim_{X \to A} E(X) = f(A)$

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Question # 1 of 2

A function may be ..... at a point
X_0

even if its first partial derivatives are not continuous at
X_0

.

Select the correct option

◉	differentiable
○	None of these
○	piecewise convergent
○	uniformly convergent

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ment Quiz

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uestion # 2 of 2

In $\mathbb{R}^2$, if the function of one variable is differentiable at $x = \alpha$, then the curve $y = f(x)$ is approximated by $f(\alpha) + f'(\alpha)(x - \alpha)$ so that - -

elect the correct option

○	$\lim_{x \to \alpha} \dfrac{f(x) + [f(\alpha) - f'(\alpha)(x - \alpha)]}{x - \alpha} = 0$
○	$\lim_{x \to \alpha} \dfrac{f(x) - [f(\alpha) + f'(\alpha)(x - \alpha)]}{x - \alpha} = 1$
○	$\lim_{x \to \alpha} \dfrac{f(x) + [f(\alpha) - f'(\alpha)(x - \alpha)]}{x - \alpha} = 1$
◉	$\lim_{x \to \alpha} \dfrac{f(x) - [f(\alpha) + f'(\alpha)(x - \alpha)]}{x - \alpha} = 0$

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Question # 2 of 2

Equation of tangent plane to $z = xy\sin(xy)$ at $\left(1, \frac{\pi}{2}\right)$ is -

Select the correct option

- $z = -\frac{\pi}{2}x + y + \frac{\pi}{2}$
- $z = \frac{\pi}{2}x - y + \frac{\pi}{2}$
- $z = \frac{\pi}{2}x + y - \frac{\pi}{2}$
- $z = \frac{\pi}{2}x - y - \frac{\pi}{2}$

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Question # 2 of 2

Equation of tangent plane to $z = 2x + 3y - 1$ at $(1, -1)$ is - -

Select the correct option

○	$z = 2x - 3y + 1$
○	$z = 2x - 3y - 1$
○	$z = -2x + 3y - 1$
●	$z = 2x + 3y - 1$

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ost-Assessment

Question # 2 of 2

In $\mathbb{R}^3$, the equation of tangent plane to the surface $z = x^2 + y^2 - 1$ at $(-1, 1)$ is - -

Select the correct option

○	$z = -2x + 2y + 3$
○	$z = 2x - 2y + 3$
●	$z = -2x + 2y - 3$
○	$z = 2x + 2y - 3$

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uestion # 1 of 2 **Time**

In $\mathbb{R}^3$, if the function of two variable is differentiable at $(x,y)=(\alpha,\beta)$, then the curve $z=f(x,y)$ is approximated by $f(\alpha,\beta)+f_x(\alpha,\beta)(x-\alpha)+f_y(\alpha,\beta)(y-\beta)$ such that;

lect the correct option

Reload Ma

- $\lim\limits_{(x,y)\to(\alpha,\beta)} \dfrac{f(x,y)-[f(\alpha,\beta)+f_x(\alpha,\beta)(x-\alpha)+f_y(\alpha,\beta)(y-\beta)]}{\sqrt{(x-\alpha)^2+(y-\beta)^2}}=0$
- $\lim\limits_{(x,y)\to(\alpha,\beta)} \dfrac{f(x,y)-[f(\alpha,\beta)+f_x(\alpha,\beta)(x-\alpha)+f_y(\alpha,\beta)(y-\beta)]}{\sqrt{f_x{}^2+f_y{}^2}}=1$
- $\lim\limits_{(x,y)\to(\alpha,\beta)} \dfrac{f(x,y)-[f(\alpha,\beta)+f_x(\alpha,\beta)(x-\alpha)+f_y(\alpha,\beta)(y-\beta)]}{\sqrt{(x-\alpha)^2+(y-\beta)^2}}=1$
- $\lim\limits_{(x,y)\to(\alpha,\beta)} \dfrac{f(x,y)-[f(\alpha,\beta)+f_x(\alpha,\beta)(x-\alpha)+f_y(\alpha,\beta)(y-\beta)]}{\sqrt{f_x{}^2+f_y{}^2}}=0$

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sment Quiz

ost-Assessment

Question # 1 of 2 **Time**

A point on the graph of a function at which the tangent line is parallel to x-axis is called the ...

Select the correct option

- None of these
- critical point
- point of inflection
- point of relative extrema

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Question # 2 of 2 — Time Left: 67 sec(s

In $\mathbb{R}^n$, a function f is said to have the local maximum point at $X = A$, if the following inequlity holds $\forall\, X \in D_f \cap S_r(A)$

Select the correct option

Reload Math Equations

- $f(X) \leqslant f(A)$
- $f(A) \leqslant f(X)$
- $f(X) < f(A)$
- $f(A) < f(X)$

Click to Save Answer & Move to Next Question

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Pre-Assessment

Question # 1 of 2

A tangent plane to the surface
$z = f(x, y)$

is always the limit of the ----------

Select the correct option

○	normal plane
○	tangent line
●	secant plane
○	None of these

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Pre-Assessment

Question # 2 of 2

$|\bar{x}|$ is the distance between x and _____.

Select the correct option

○	identity
○	$\bar{y}$
○	x
●	the origin

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Pre-Assessment

Question # 1 of 2

In R^4 let $\bar{x} = (1, -3, 5, 6)$ and $\bar{y} = \left(2, -4, 3, \frac{1}{2}\right)$, $\bar{x} + \bar{y} =$ __

Select the correct option

◉	$(3, -7, 8, 6.5)$
○	$\left(3, -7, 8, \frac{7}{2}\right)$
○	$(3, 7, 8, 6.5)$
○	$(2, -7, 8, 6)$

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Assessment Quiz

Pre-Assessment

Question # 1 of 2

The vector sum of $\bar{x} = (x_1, x_2, x_3, x_4, x_5, x_6)$ and $\bar{y} = (y_1, y_2, y_3, y_4, y_5, y_6)$ is $\bar{x} + \bar{y} =$ _

Select the correct option

- ○ $(x_6 + y_6, x_5 + y_5, \cdots, x_1 + y_1)$
- ○ $(x_1 + y_1 + x_2 + y_2 + \cdots + x_6 + y_6)$
- ◉ $(x_1 + y_1, x_2 + y_2, \cdots, x_6 + y_6)$
- ○ $(x_1 + y_1.\ x_2 + y_2.\cdots.x_6 + y_6)$

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Pre-Assessment

Question # 2 of 2

If $\bar{x}$ and $\bar{y}$ are in R^n, then _____ is called triangle inequality.

Select the correct option

- ○ $|\bar{x}+\bar{y}| > |\bar{x}|+|\bar{y}|$
- ◉ $|\bar{x}+\bar{y}| \leqslant |\bar{x}|+|\bar{y}|$
- ○ $|\bar{x}+\bar{y}| \geqslant |\bar{x}|+|\bar{y}|$
- ○ $|\bar{x}+\bar{y}| < |\bar{x}|+|\bar{y}|$

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Pre-Assessment

Question # 1 of 2

If $\bar{x}$ and $\bar{z}$ are in R^n, then _____ $||\bar{x}| - |\bar{z}||$.

Select the correct option

○	$\|\bar{x} - \bar{z}\| <$
○	$\|\bar{x} - \bar{z}\| \leqslant$
○	$\|\bar{x} - \bar{z}\| >$
●	$\|\bar{x} - \bar{z}\| \geqslant$

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essment Quiz

Pre-Assessment

Question # 2 of 2

The equation of a line through a point $X_0 = (x_0;\ y_0;\ z_0)$ in R^3 can be written parametrically as _ $-\infty < t < \infty$ where $u_1, u_2, u_3 \neq 0$.

Select the correct option

Rel

○	$z = z_0 + u_3 t$
○	$x = x_0 + u_1 t$
○	$y = y_0 + u_2 t$
◉	all of above

Click to Save Answer & Move

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Pre-Assessment

Question # 1 of 2 **Time Left: 79 s**

Topological Properties of $\mathbb{R}^n$ can be described in terms of special class of subsets, called- - - - - - - - - in $\mathbb{R}^n$.

Select the correct option

Reload Math Equatio

- ○ closed sets
- ○ open sets
- ◉ all are equivalent
- ○ neighborhoods

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Pre-Assessment

Question # 2 of 2 Ti

If the points 0, X, Y and $X+Y$ are joined to form a parallelogram in $\mathbb{R}^n$ of sides $|X|$ and $|Y|$, then its diagonal is represented by - - - - - - - - .

Select the correct option

Reload

○	$\|X\|+\|Y\|$
○	$\|\|X\|-\|Y\|\|$
◉	$\|X+Y\|$
○	$\|X\|-\|Y\|$

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essment Quiz

Question # 2 of 2

In the Schwarz's Inequality; $|X.Y| \leqslant |X|\,|Y|\,, \forall X, Y$ in $\mathbb{R}^n$, the equlaity holds for - - - -

Select the correct option

- $Y = tX,\ t \in \mathbb{R}$
- $X = Y$
- $X = tY,\ t \in \mathbb{R}$
- All above are valid

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Pre-Assessment

Question # 2 of 2

The length of the vector $\bar{x} = (x_1, x_2, 3, x_4, 5)$ is $|\bar{x}| =$ _

Select the correct option

○	$\left(x_1^2 + x_2^2 + 9 + x_4^2 + 25\right)$
○	$\left(x_1^2 + x_2^3 + 3^4 + x_4^5 + 5^6\right)^{\frac{1}{2}}$
○	$\left(x_1^2, x_2^2, 9, x_4^2, 25\right)^{\frac{1}{2}}$
◉	$\left(x_1^2 + x_2^2 + x_4^2 + 34\right)^{\frac{1}{2}}$

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Pre-Assessment

Question # 2 of 2 **Time Left:**

Both $X = X_0 + sV$ and $X = X_0 + tU,\ -\infty < s, t < \infty$ represent the same line through X_0 in $\mathbb{R}^n$, if - -

Select the correct option

Reload Math Ec

- $U = \beta V,\ \beta \neq 0$
- All above are equivalent
- $sV = tU,\ s \neq 0$ or $t \neq 0$
- $V = \alpha U, \alpha \neq 0$

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Question # 2 of 2 Time Left: 83 sec(s)

If partial derivatives $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ are identically zero in an open region S of R^n, then f is _____ in S.

Select the correct option

Reload Math Equations

- [ ] None of these
- [ ] Bounded
- [x] Constant
- [ ] Unbounded

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Question # 2 of 2 **Time Left: 83 sec(s**

Suppose that f and its partial derivatives of order $\leq k$ are differentiable at $\mathbf{X}_0$ and $\mathbf{X}$ in $\mathbb{R}^n$ and on the line segment L connecting them.

Then

Select the correct option

Reload Math Equations

- $f(\mathbf{X}) = \sum_{r=0}^{k} \frac{1}{r!}(d^{(r)}_{\mathbf{X}_0} f)(\mathbf{X} - \mathbf{X}_0) + \frac{1}{(k+1)!}(d^{(k+1)}_{\widetilde{\mathbf{X}}} f)(\mathbf{X} - \mathbf{X}_0)$ for some $\widetilde{\mathbf{X}}$ on L distinct from $\mathbf{X}_0$ and $\mathbf{X}$.
- $f(\mathbf{X}) = \sum_{r=0}^{k} \frac{1}{r!}(d^{(r)}_{\mathbf{X}_0} f)(\mathbf{X} - \mathbf{X})$

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uestion # 2 of 2 Time Left: 81 sec(s)

Over the complete domain we have taken a line segment the function of several variables is differentiable.

elect the correct option

Reload Math Equations

- (●) True
- ( ) False

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Question # 2 of 2 Time L

Taylor's theorem states for function of n variables f and its partial derivatives are differentiable at X_0 and X in R^n for order_____.

Select the correct option

Reload Math

- $< k$
- $\leq k$
- $\geq k$
- $> k$

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Post-Assessment

Question # 1 of 2

The expression $(d^{(0)}_{\mathbf{X}_0} f) =$

Select the correct option

- (●) $f(\mathbf{X}_0)$.
- ( ) $d_{\mathbf{X}_0} f$.

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Question # 2 of 2

The function $F(x) = \int_a^x f(t)\,dt$ is

Select the correct option

○	undefined on $[a, b)$.
◉	nondecreasing on $[a, b)$.
○	constant on $[a, b)$.
○	decreasing on $[a, b)$.

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Post-Assessment

Question # 2 of 2

If f and g are locally integrable on [a,b) and $0 \le f(x) \le g(x)$, $a \le x < b$, then

Select the correct option

- $\int_a^b f(x)\,dx < \infty, if \int_a^b g(x)\,dx < \infty$
- $\int_a^b f(x)\,dx = 0, if \int_a^b g(x)\,dx < \infty$

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ost-Assessment

Question # 1 of 2

If f is nonnegative and locally integrable on $[a, b)$, then $\int_a^b f(x)\,dx$ converges

Select the correct option

- (●) if the function $F(x) = \int_a^x f(t)\,dt$ is bounded on $[a, b)$, and $\int_a^b f(x)\,dx = \infty$.
- ( ) if $\int_a^b f(x)\,dx = 0$.
- ( ) if the function $F(x) = \int_a^x f(t)\,dt$ is bounded on $[a, b)$.
- ( ) if $\int_a^b f(x)\,dx = \infty$.

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Post-Assessment

Question # 2 of 2

The function $f(x) = (1 - x)^{-p}$ is locally integrable on

Select the correct option

○	$[-\infty, 0)$.
○	$[0, \pi)$.
○	$[0, \infty)$.
●	$[0, 1)$.

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Post-Assessment

Question # 2 of 2

The function $f(x) = x^{-p}$ is locally integrable on

Select the correct option

○	$[0, 1)$.
○	$[0, \infty)$.
●	$[1, \infty)$.
○	$[-\infty, \infty)$.

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Post-Assessment

Question # 2 of 2

The function $h(x) = \sqrt{x}$ is locally integrable on

Select the correct option

- (●) $[0, \infty)$.
- ( ) $[0, 1)$.

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Post-Assessment

Question # 1 of 2

$f(x) = \sin x$ is locally integrable on

Select the correct option

◉	$(-\infty, \infty)$.
○	$(0, \infty)$.
○	$(-\infty, 0)$.
○	$(-\infty, 1)$.

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Pre-Assessment

Question # 1 of 2

The expression $(d^{(0)}_{\mathbf{X}_0} f) =$

Select the correct option

- ○ $d_{\mathbf{X}_0} f.$
- ● $f(\mathbf{X}_0).$

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ment Quiz

e-Assessment

Question # 2 of 2 — Time Left: 84 sec(s)

If partial derivatives $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ are identically zero in an open region S of R^n, then f is _____ in S.

Select the correct option

Reload Math Equations

- [ ] None of these
- [x] Constant
- [ ] Unbounded
- [ ] Bounded

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Question # 1 of 2

For $f(x, y) = e^{ax^2+by^2}$.

Select the correct option

○	$f_x(0, 0) = 0.$
○	$(0, 0)$ is a critical point of f.
●	all options are correct.
○	$f_y(0, 0) = 0.$

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e-Assessment

uestion # 1 of 2 **Time Left: 6**

By analogy with the situation for functions of one variable, we define the kth Taylor polynomial of f about X_0 by

elect the correct option

Reload Math Equ

- (●) $T_k(\mathbf{X}) = \sum_{r=0}^{k} \frac{1}{r!}(d^{(r)}_{\mathbf{X}_0} f)(\mathbf{X} - \mathbf{X}_0).$
- ( ) $T_k(\mathbf{X}) = (d^{(r)}_{\mathbf{X}_0} f)(\mathbf{X} - \mathbf{X}_0).$

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Question # 1 of 2 Time Lef

Taylor's theorem states for function of n variables f and its partial derivatives are differentiable at X_0 and X in R^n for order_____.

Select the correct option

Reload Math E

- $> k$
- $\geq k$
- $< k$
- $\leq k$

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Question # 1 of 2 **Time Left: 80 sec(s)**

Suppose that f and its partial derivatives of order $\leq k$ are differentiable at $\mathbf{X}_0$ and $\mathbf{X}$ in $\mathbb{R}^n$ and on the line segment L connecting them.

Then

Select the correct option

Reload Math Equations

- $f(\mathbf{X}) = \sum_{r=0}^{k} \frac{1}{r!}(d_{\mathbf{X}_0}^{(r)} f)(\mathbf{X} - \mathbf{X}_0) + \frac{1}{(k+1)!}(d_{\overline{\mathbf{X}}}^{(k+1)} f)(\mathbf{X} - \mathbf{X}_0)$ for some $\overline{\mathbf{X}}$ on L distinct from $\mathbf{X}_0$ and $\mathbf{X}$.
- $f(\mathbf{X}) = \sum_{r=0}^{k} \frac{1}{r!}(d_{\mathbf{X}_0}^{(r)} f)(\mathbf{X} - \mathbf{X})$

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Question # 2 of 2 Time Left: 79 sec(s)

Over the complete domain we have taken a line segment the function of several variables is differentiable.

Select the correct option

Reload Math Equations

- (●) True
- ( ) False

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Pre-Assessment

Question # 1 of 2

For $f(x, y) = e^{ax^2+by^2}$.

Select the correct option

○	$f_x(0, 0) = 0.$
○	$(0, 0)$ is a critical point of f.
○	$f_y(0, 0) = 0.$
◉	all options are correct.

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Question # 1 of 1

The lower sum of a function f over the partition P is _

Select the correct option

○	$\sum_{j=1}^{k} f(X_j)V(R_j)$
◉	$\sum_{j=1}^{k} m_j V(R_j)$
○	$\sum_{j=1}^{k} M_j V(R_j)$
○	All of these

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Question # 1 of 1

The upper sum of a function f over the partition P is _

Select the correct option

○	$\sum_{j=1}^{k} m_j V(R_j)$
●	$\sum_{j=1}^{k} M_j V(R_j)$
○	$\sum_{j=1}^{\infty} m_j V(R_j)$
○	$\sum_{j=1}^{\infty} M_j V(R_j)$

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ssment Quiz

Post-Assessment

Question # 1 of 1 **Time Left**

If f is unbounded on the nondegenerate rectangle R in R_n then f is integrable on R.

Select the correct option

Reload Math Ec

- ○ True
- ◉ False

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Question # 2 of 2

The Riemann sum of a function f over the partition P of R is __

Select the correct option

- ○ $\sum_{j=1}^{k} m_j V(R_j)$
- ◉ $\sum_{j=1}^{k} f(X_j)V(R_j)$
- ○ $\sum_{j=1}^{\infty} M_j V(R_j)$
- ○ $\sum_{j=1}^{\infty} f(X_j)V(R_j)$

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Post-Assessment

Question # 2 of 2

Dirichlet's test is useful only for _____ function.

Select the correct option

- ○ Bounded
- ◉ Continuous and Oscillatory
- ○ Continuous
- ○ Oscillatory

C

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Post-Assessment

Question # 2 of 2 **Time Left: 84 s**

The content of R is $V(R) = (b_1 - a_1)(b_2 - a_2)\ldots(b_n - a_n)$ if $V(R)=0$ then R is called _____.

Select the correct option

Reload Math Equatio

- ○ Nondegenerate
- ◉ Degenerate
- ○ a & c
- ○ coordinate cube

Click to Save Answer & Move to Next Question

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Post-Assessment

Question # 1 of 2 **Time Left: 81 sec**

If f is defined on [a, b) and changes sign on every such subinterval $[a_1, b)$ of [a, b) is called _____.

Select the correct option

Reload Math Equations

- (●) Oscillatory
- ( ) None
- ( ) Nonoscillatory
- ( ) Bounded

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Post-Assessment

Question # 2 of 2 **Time Left: 75 se**

The content of R is $V(R) = (b_1 - a_1)(b_2 - a_2)\ldots(b_n - a_n)$ if $V(R)=0$ then R is called _____.

Select the correct option

Reload Math Equation

○	Nondegenerate
○	coordinate cube
○	a & c
◉	Degenerate

Click to Save Answer & Move to Next Question

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Question # 1 of 2 Time L

An oscillatory function may be integrable, but not absolutely integrable on $[a, b)$.

Select the correct option Reload Math

- False
- True

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Question # 1 of 2 Time Left: 72 sec(

If f is defined on [a, b) and changes sign on every such subinterval $[a_1, b)$ of [a, b) is called _____.

Select the correct option

Reload Math Equations

- Bounded
- None
- Nonoscillatory
- Oscillatory

Click to Save Answer & Move to Next Question

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Question # 2 of 2

The integral $\int_1^{\infty} x^q \sin x dx$ diverges if __

Select the correct option

○	q = 1
○	q < 0
●	q > 0
○	q = 0

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ssment Quiz

Post-Assessment

Question # 1 of 2

Dirichlet's test is useful only for _____ function.

Select the correct option

- ○ Bounded
- ○ Oscillatory
- ◉ Continuous and Oscillatory
- ○ Continuous

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Question # 2 of 2

$f(x) = \sin x$ is locally integrable on

Select the correct option

○	$(-\infty, 0)$.
○	$(0, \infty)$.
●	$(-\infty, \infty)$.
○	$(-\infty, 1)$.

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e-Assessment

uestion # 1 of 2

If f is nonnegative and locally integrable on $[a, b)$, then $\int_a^b f(x)\,dx$ converges

elect the correct option

○	if $\int_a^b f(x)\,dx = \infty$.
○	if the function $F(x) = \int_a^x f(t)\,dt$ is bounded on $[a, b)$.
○	if $\int_a^b f(x)\,dx = 0$.
◉	if the function $F(x) = \int_a^x f(t)\,dt$ is bounded on $[a, b)$, and $\int_a^b f(x)\,dx = \infty$.

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Question # 1 of 2

The function $F(x) = \int_a^x f(t)\,dt$ is

Select the correct option

◉	nondecreasing on $[a, b)$.
○	undefined on $[a, b)$.
○	decreasing on $[a, b)$.
○	constant on $[a, b)$.

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